BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SWAT

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: 112301

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Minutes | **Maximum Marks:** 18

Note: 1) Attempting all MCQs is compulsory. This paper along with the OMR sheet must be returned to the superintendent after due time. 2) Fill the circle A B C D, which one is correct with blue or black ball point in separate OMR Sheet like ● 3) If more than one circle in the OMR sheet is filled then no credit will be given to such answer.

Q1. Difference in magnitude of a property in different directions is known as

(A) Cleavage
(B) Isotropy
(C) Polymor phism
(D) Anisotropy

Q2. An orbital that is filled first with electrons, is

(A) 3d
(B) 4P
(C) 5S
(D) 5P

Q3. According to lowry-Bronsted concept, water is in nature

(A) Acidic
(B) Basic
(C) Neutral
(D) Amphoteric

Q4. The solubility of gases decrease with increase in

(A) Pressure
(B) Volume
(C) Temperature
(D) All of these

Q5. When reactants and products are liquids or solids then

(A) $\Delta H=0$
(B) $\Delta H=\Delta E$
(C) $\Delta H=\Delta E+P\Delta V$
(D) $\Delta H=P\Delta V$

Q6. At same temperature and pressure, which gas will diffuse most slowly

(A) CO_2
(B) N_2
(C) NH_3
(D) CH_4

Q7. Bond energy of a covalent bond decreases with increase in

(A) Polarity
(B) Multiplicity
(C) Size of atoms
(D) Ionic character

Q8. At equilibrium state, the pressures/concentrations of reactants and products

(A) Increase
(B) Decrease
(C) Equal
(D) Constant

Q9. When equal moles of hydrogen and oxygen react to form water, then the limiting reactant is.

(A) H_2
(B) O_2
(C) H_2O
(D) None

Q10. Which element is liberated at cathode during electrolysis of aqueous NaCl solution

(A) Sodium
(B) Hydrogen
(C) Chlorine
(D) Oxygen

Q11. One molal aqueous solution of which salt has the highest boiling point

(A) NaCl
(B) $CaCl_2$
(C) $MgCl_2$
(D) $AlCl_3$

Q12. According to Hess's law, enthalpy change for a cyclic process is

(A) Zero
(B) Low
(C) Moderate
(D) High

Q13. Order of a reaction may be

(A) Zero
(B) Whole number
(C) Fractional
(D) All of these

Q14. A solute in solution neither precipitates nor further dissolves when

(A) IP > K_{sp}
(B) IP < K_{sp}
(C) IP = K_{sp}
(D) IP ≥ K_{sp}

Q15. Which buffer solution changes the added strong acid into weak acid

(A) Neutral buffer
(B) Basic buffer
(C) Acid buffer
(D) None of these

Q16. The geometry of a triatomic molecule with zero dipole moment is

(A) Planar trigonal
(B) tetrahedral
(C) Angular
(D) Linear

Q17. An increase in internal energy of a system may cause

(A) Chemical change
(B) Phase change
(C) Temperature rise
(D) All of these

Q18. Oxidation number of hydrogen in metal hydrides is

(A) -1
(B) [illegible in source scan]
(C) +2
(D) Zero

SECTION "B"

Time Allowed: 2 hours and 40 minutes | **Maximum Marks:** 40

QII. Attempt any Ten Parts out of the following. Each Part carries equal marks.

(i) How soaps and detergents increase the cleansing action of water?

(ii) Calculate the wave number of a spectral line of electron when jumps from n=4 to n=2

(iii) Why balanced chemical equations are used in stoichiometric problems?

(iv) How dipole moment helps to determine the structures of covalent molecules?

(v) Calculate pH and pOH of 0.0005M $Ca(OH)_2$ solution.

(vi) Why rapid expansion of gases causes cooling?

(vii) What is meant by specific rate constant? Give its units for 1st and 2nd order reactions.

(viii) What is meant by state and state function?

(ix) How corrosion of metals can be prevented?

(x) What are the four types of crystalline solids on the basis of nature of bonding?

(xi) How pressure change affects the equilibrium state of a chemical reaction?

(xii) Calculate the mole fraction of NaOH when 40g NaOH is dissolved in 54g H_2O.

(xiii) What is meant by Joule Thomson effect?

SECTION "C"

Maximum Marks: 27

Note: Attempt any Three questions of the following. Each question carries equal Marks.

QIII.

(a) [5] State and explain law of mass action.

(b) [4] Why earthen pots keep water cool in hot summer?

QIV.

(a) [5] What are Quantum numbers? Explain Principle and Azimuthal Quantum number in detail.

(b) [4] How limiting reactant can be identified in chemical reaction?

QV.

(a) [5] Define hybridization? Discuss sp^3 and sp^2 hybridization in detail.

(b) [4] How buffer solution keeps its pH constant?

QVI.

(a) [4] What is meant by molarity and molality of solution?

(b) [5] What is electrode potential? How it can be measured?